



HACKYEAH 2026 · KRAKÓW · OPEN TASK: SPORT & HEALTHCARE

JustMate

Meet for real.

A health & wellbeing app for the loneliness epidemic. JustMate gets lonely people out of the door and in front of someone compatible — faceless, mutual and on foot. Right now, or at a small plan the app makes for you.

DOCUMENT

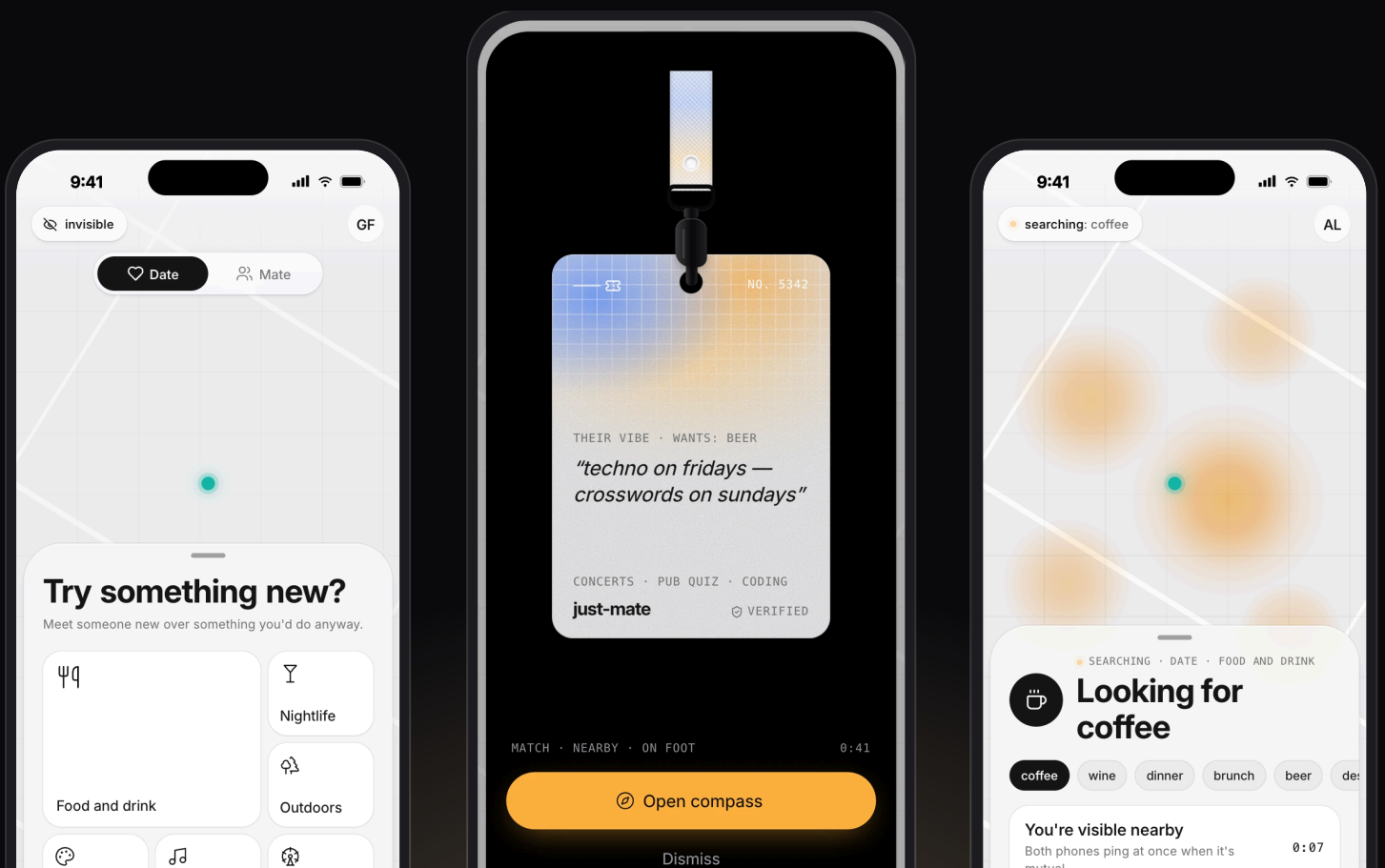
Product & technical whitepaper
The jury's reference to the project

THEME

Health & wellbeing: loneliness
Meet, move, look up — not therapy

REPOSITORY

github.com/uteg-labs/just-mate
Version of 4 October 2026



Apps solved matching. Nobody solved the door.

Loneliness is a health risk on a par with smoking. A lonely person rarely goes out on a whim: they need a reason, a time, a place, and proof that someone will be there. JustMate gives them exactly that — and the walk there.

1 in 6

people worldwide are affected by loneliness

WHO, 2025

871k

deaths a year are linked to loneliness

WHO, 2025

2x

lonely people are twice as likely to become depressed

WHO, 2025

~50 h

together to turn an acquaintance into a casual friend

Hall, 2018

THE PROBLEM

Three barriers keep a lonely person at home

Initiative — organising anything feels impossible. **The empty table** — "what if nobody comes?" **Judgment** — photo-first apps make meeting an audition.

OUR ANSWER

One faceless profile, two speeds

Plan — the app proposes a plan with someone compatible, at a place halfway. Tap accept; it's on only if they accept too. **Now** — already out? Two compatible people nearby are pinged at once and a compass walks them together.

THE CONSTITUTION

Three rules for every feature

No faces — nobody is judged on a photo. **No chat** — every flow ends face to face. **No people pins** — a person is never a point on a map; only places are.

WHAT WE BUILT IN 24 HOURS

- **Native iOS & Android app** (Expo dev-client): faceless onboarding with a live LLM interview, map, Plans, match card, compass, post-meet; English, Polish & Slovak.
- **Bun + Elysia server**: accounts and profiles in PostgreSQL, one WebSocket per phone with a 1-second matching loop, a compass relay that sends a bearing — never a position — and the Plans engine over seeded public venues.
- **A shared, typed protocol** — written before code — used by both app and server; **118 automated server tests**.
- **A learned matching model** (PyTorch → ONNX) trained on 25,000 synthetic profiles alongside the explainable score the demo runs on.

NUMBERS THAT DEFINE THE PRODUCT

0

positions ever stored

400m

walking-distance match gate

10 min

compass window, then it expires

~21

partner venues cover a city's running costs (est.)

CONTENTS • EVERY SENTENCE DESCRIBES WHAT THE BUILD DOES, OR IS LABELLED

PRODUCTION PATH

01 The problem & health case

02 The product

03 Who it's for

04 The Plan loop

05 The Now loop

06 Experience & design

07 Safety & privacy

08 Architecture

09 Matching & AI

10 Engineering

11 Business model

12 Launch & competition

13 Metrics & roadmap

14 Honesty & sources

Loneliness, and the door

Loneliness touches about one in six people and is linked to an estimated 871,000 deaths a year. It is highest among young people — 17–21% of 13–29-year-olds — and lonely people are twice as likely to become depressed (WHO Commission on Social Connection, June 2025).

Matching is solved. Meeting is broken.

It breaks hardest for a lonely, maybe low person. Existing apps monetise attention spent *on the phone*: their incentive is the opposite of getting you outside. For someone who already finds it hard to go out, three barriers stand between them and a real conversation:

- 1 Initiative.** Low mood means low energy. Organising something — even choosing between options — is too much. "Find people now" assumes you are already out and already brave.
- 2 The empty table.** "What if nobody comes? What if I'm stood up?" One no-show confirms the story *nobody wants me*.
- 3 Judgment.** Photo-first apps turn meeting into an audition; chat turns it into weeks of performance before anything real happens.

"Apps solved matching. Nobody solved the door."

The product's job is not the first meeting, but the second, the third and the tenth: friendship takes roughly 50 hours together to go from acquaintance to casual friend (Hall, *Journal of Social and Personal Relationships*, 2018).

DESIGN CONSEQUENCE

Every mechanic in JustMate removes one of the three barriers. The phone's job ends where the conversation starts — **the interface is the street.**

Why existing formats fail

FORMAT	FAILURE MODE FOR A LONELY PERSON
Swipe apps (Tinder, Bumble)	Photo-first judgment; chat purgatory; meetings are the exception, not the goal.
Friend-matched blind dating (BlindMate)	Photos hidden only until you chat; your friends must swipe for you — the circle a lonely person lacks. Still chat-first, dating only.
Scheduled dating (Breeze)	Real meetings, but planned hours or days ahead — and dating only.
Stranger dinners (Timeleft)	Proves strangers show up to a fixed plan, but on fixed weekly slots, paid per seat, anywhere in the city.
AI-matched groups (Pie)	Small groups matched by a quiz, then a group chat before the event — the chat gate again.
Proximity dating (happn)	Retrospective ("who you passed 7 days ago"), photo-first, chat-gated.
Ambient proximity (Highlight, Sonar)	Always-on location, battery drain, empty rooms, safety gaps. Both are gone.
Activity platforms (Meetup)	Needs an organiser and a group; big rooms are hard for the shy; nothing for "a beer buddy now".

One finding we build on: Timeleft's co-founder observed that "the more information you give, the more people don't show up". A faceless badge is the right amount of information.

Connection is health. It starts at the door.

JustMate is a health & wellbeing product for the loneliness epidemic. It treats nothing; it changes what people do — **meet** someone in person, **move** to get there, **look up** from the phone while they talk — and come back again.

+50%

odds of survival with strong social ties — an effect comparable with smoking

Holt-Lunstad, 2010

5

risks it raises: stroke, heart disease, diabetes, cognitive decline, early death

WHO, 2025

2×

as likely to become depressed when lonely

WHO, 2025

31%

of adults — 1.8 billion — don't get enough physical activity

WHO, 2024

WHO asks every country to make social connection a public-health priority. The everyday steps it names read like our spec — *"putting away one's phone to be fully present in conversation", "joining a local group"* — with cafés and parks as social infrastructure and walking as active transport.

Four pathways from an app to wellbeing

PATHWAY	WHAT JUSTMATE DOES	WHAT CHANGES
Meet · connection	Every flow ends face to face: a plan the app proposes, or a ping when you're already out. No chat to get stuck in; built for the second meeting (friendship ≈ 50 h together).	Regular in-person contact — the protective factor in the studies above
Move · activity	Every meeting starts on foot: Now matches only within a 5-, 10- or 15-minute walk; plans sit at a public place halfway. 16 of the 44 Mate activities are sport or outdoors — running, climbing, padel, football, yoga, hiking, cycling, a walk.	Light activity on the way, and a partner for sport today
Look up · screens	No feed, chat, likes or streaks — nothing to scroll. Time in the app is an anti-metric (ch. 13).	Time in company with the phone away — WHO's own example
Small step · pressure	A faceless badge, a mutual yes, a fixed time and a public place: one small, scheduled step, as in behavioural activation (ch. 04). Declines are silent.	Going out becomes possible on a low-energy day

How we will show it works — without health data

IN THE PRODUCT • BEHAVIOUR, NEVER HEALTH

- **Real-world meetings** per active user per week; repeat meetings with the same person or group
- **Movement:** sport or outdoor share of meetings; minutes walked to meet **M1**
- **Time in app** should not grow

OUTCOME STUDY • RESEARCH PARTNER M1

Opt-in and anonymous: the 3-item UCLA Loneliness Scale and the WHO-5 Well-Being Index at sign-up and after 8 weeks, run by a Kraków university partner under its own consent and ethics approval. JustMate sees aggregates only.

Wellbeing, not medicine. No health claims (EU MDR), no health or mood data (GDPR Art. 9), no guilt mechanics (ch. 02). Psychologists, counsellors and GPs can recommend JustMate as a next step between sessions (social prescribing, ch. 12).

Three rules. Two speeds.

One faceless profile, two ways out of the door. **Plan** gives you a reason to go, with time to get ready. **Now** is serendipity for when you are already out. Both end the same way: two people talking face to face, first names unlocked, and one tap for *same again next week?*

The constitution

Every feature decision is tested against three rules. Anything that violates one is out by definition.



No faces

No profile photos anywhere. Attraction is a private compatibility signal, never a picture to be judged. Kills the appearance economy and the anxiety of profile curation. The only faces in the app are generated samples used to learn taste — never a user.



No chat

There is no messaging. The success event is a real conversation, so every feature points at it — the match output is *legs, not thumbs*. Plans coordinate with fixed status chips ("on my way", "5 min late", "I'm here · by the window"), never free text.



No people pins

A person's location is never a point — only anonymous zones with an aggregate warmth. Nobody can be found, followed or looked up. Places can be pinned: a plan's venue is public. People never are. The rule is enforced by the protocol, not just the UI.

Two speeds



PLAN · WHEN YOU NEED A REASON TO GO

The app proposes, you only say yes

A plan with a compatible person — *board games · Sunday 15:00 · Meeple Café, halfway for you both* — with their vibe badge. Tap **Accept plan**; it's on only when they accept too. Or send your own invitation — you don't pick who. In Mate, group plans for 3–4.

Removes: initiative · the empty table · judgment



NOW · WHEN YOU'RE ALREADY OUT

Both phones ping at the same moment

Pick what you're up for — a beer, a coffee, a game — and tap **Find people**. When a compatible person who wants the same thing is within a 5-minute walk, both get a match card at once. If both accept, a hot/cold compass guides them to each other within ten minutes.

Wedge: consented serendipity — real-time, mutual, faceless, on foot

THE MENTAL-HEALTH LINE

Not therapy. A reason to go out.

No health claims. Software that treats a condition is a medical device under the EU MDR (2017/745). We never say JustMate treats anything.

No mood or health data. That is special-category data under GDPR Art. 9. We never ask how you feel.

No guilt mechanics. No streaks to lose, no "you haven't gone out in 7 days". Declines are silent.

Signposting. Settings carry "Need to talk to someone?" with local support lines. The app never plays counsellor.

The quietly lonely in a city

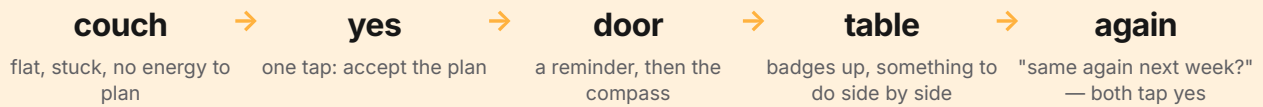
Primary: 18–35-year-olds who moved for work or study, remote workers, people after a breakup, students who haven't found their people yet. They can walk, and cities give the density we need.

THE HERO • Tomek, 28

Moved to Kraków four months ago. Works from home. Hasn't said a word out loud all weekend.

Not "depressed" — flat and stuck. Organising anything feels impossible; a big Meetup room feels worse.

*Sunday night a plan drops in: board games, Sunday 15:00, Meeple Café, halfway for both — with a badge that reads "will lose at chess — will demand a rematch". He taps **Accept plan**: "you're both in". On the day, a reminder, then the compass, then someone at the table with a chessboard. "Same again next week?" Both tap it. A month later it's a table of four.*



The supporting cast

THE TOURIST

Lucía, 27

Alone in Kraków, out for an evening walk. Mate → beer. Two minutes later both phones ping: someone 300 m away wants a beer too. **Now loop.**

THE DATER

Ania, 24

Deleted Tinder twice; tired of photo judgment and chats that die. At a festival, compatible people can find her without knowing who she is. **Date mode.**

THE SAFETY LENS

Marta, 31

"Nobody knows where I am unless I said: now, this person, yes — and I can end it instantly." **Safety.**

THE PARTNER

Piotr, 38

Board-game café, empty tables on Tuesday evenings. Lists two slots a week; JustMate fills them. **Revenue.**

Use cases

WHO	SITUATION	LOOP	WHAT JUSTMATE DOES
Newcomer (Tomek)	Lonely, low energy, never initiates	Plan	A plan with a compatible person, halfway; he only taps Accept
Shy person	Big groups and 1:1 dates both feel like too much	Plan (3–4)	Activity-first plan: something to do side by side, not an interview
Tourist (Lucía)	Alone in a city, already out walking	Now	A ping with someone compatible within 400 m
Single (Ania)	Wants to date without photos	Now / Plan	A faceless match nearby, or a date proposal halfway
Venue (Piotr)	Empty tables off-peak	Plan	Lists slots; JustMate fills them M1

Not now: older adults — they need a phone-light channel, reached later through partners. **Age:** Date is 18+; Mate admits 16–17-year-olds, matched only with each other.

Getting out of the door

How do you get a lonely, maybe low person to actually go? Take every decision off their plate except one: **yes**. Plans meet later, same rules — no faces, no chat — and a plan is on only when the other side has said yes too.



1 • PROPOSED FOR YOU • DATE OR MATE

The app plans, you say yes

The matching model finds a strong match who is usually around when you are, picks something you both like and a public place roughly halfway. You see their vibe badge, the match, the time and the place — one tap: **Accept plan**. It confirms only if they accept too.



2 • YOUR INVITATION • DATE OR MATE

Plan for later — you don't pick who

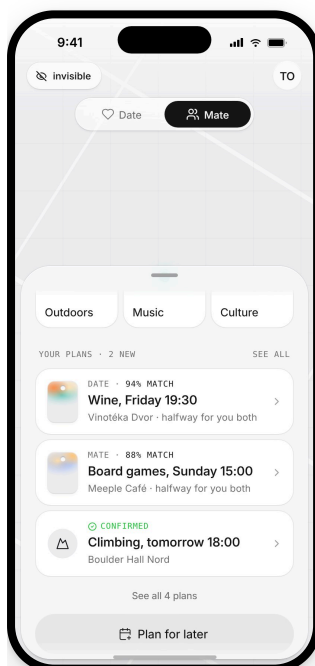
What, when (several days and times), where — send. It is offered, one at a time, to compatible people who are usually around at one of your times. The first one in picks a time and shows up as a vibe; you confirm.



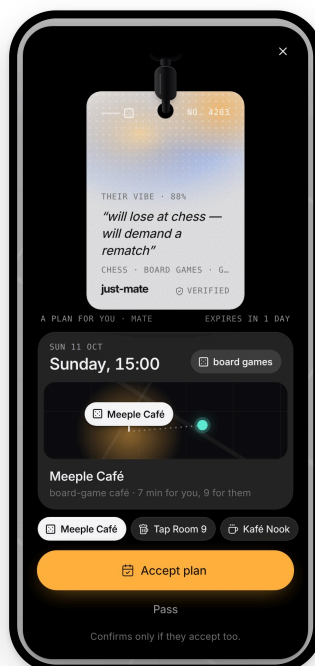
3 • GROUP PLANS • MATE, 3-4 PEOPLE

A small table, not a big room

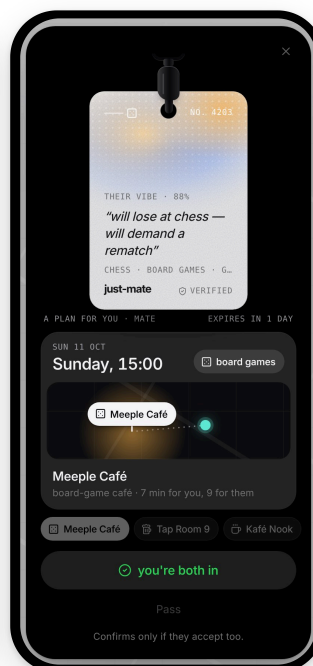
App-made plans at public venues: I'm in, "2 of 4 going" as badges, and the plan goes ahead only when the minimum is reached — on every phone at once.



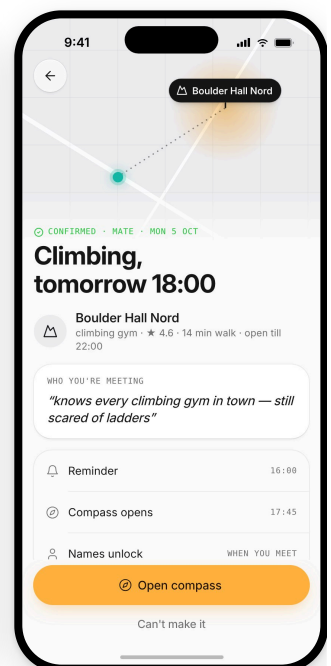
YOUR PLANS
ON THE MAP SHEET



A PLAN FOR YOU
VIBE • TIME • HALFWAY



YOU'RE BOTH IN
MUTUAL, LIKE A MATCH

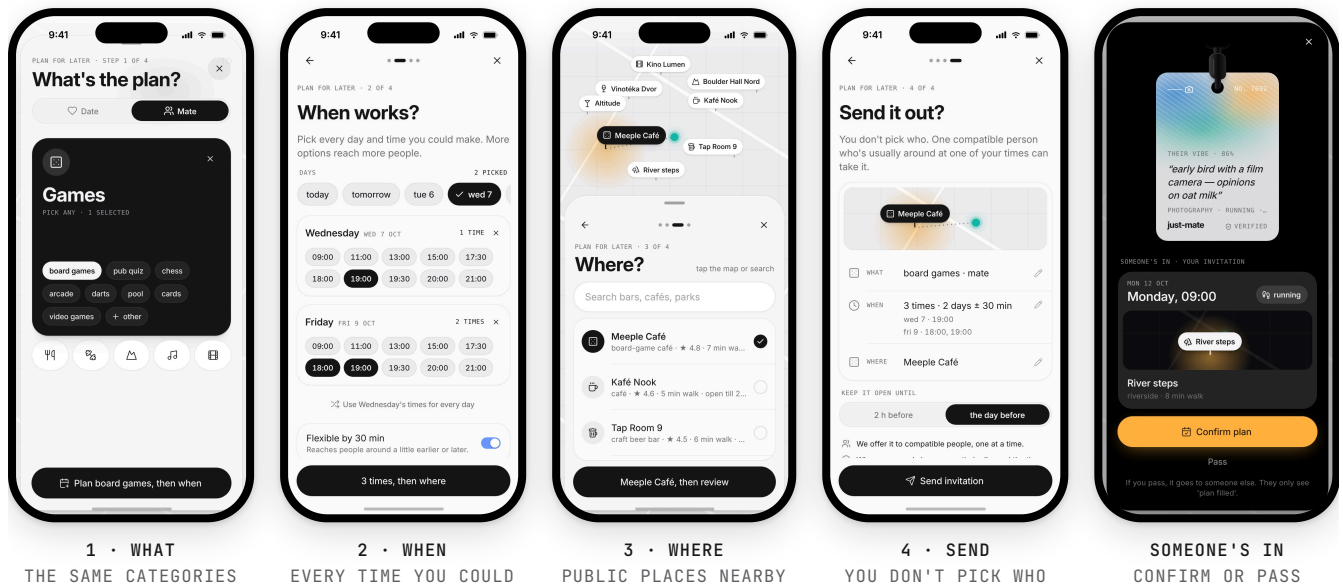


CONFIRMED
REMINDER • COMPASS

Tomek's Sunday night: a proposal drops in — *board games, Sunday 15:00, Meeple Café, 7 min for you, 9 for them* — with a badge that reads *"will lose at chess — will demand a rematch"*. He taps **Accept plan**; "waiting for them..." turns into **you're both in**. He didn't plan anything, didn't chat with anyone, and he knows someone will be there.

Plan for later: an invitation in four steps

For the days you do have the energy — or a thing you'd do anyway. The user decides *what, when and where*; JustMate decides *who*, from compatibility, and never shows faces or names.



The rules of an invitation

- **More options reach more people.** Pick several days and several times each; "flexible by 30 min" reaches people around a little earlier or later.
- **Offered one at a time** to compatible people who are usually around at one of your times. They see a vibe, a place and a time — nothing else.
- **The first one in picks a time** and appears as a vibe badge with the match. You confirm, or pass — then it goes to someone else, and they only see "plan filled".
- **Open until 2 hours or the day before**; withdraw any time.

Once a plan is confirmed

- **Reminder 2 hours before**; the plan card shows the place, the walk and *who you're meeting* — as a vibe.
- **The compass opens 15 minutes before the start**: the last hundred metres work exactly like Now — bearing and bucket, never a pin.
- **Names unlock when you meet**. Keep in touch is mutual; same again next week? is one tap.
- **Can't make it is one tap too**. The other side sees "plan cancelled" — no reason asked, no blame.

GROUP PLANS • MATE

For people for whom a 1:1 feels like too much: plans for 3–4 at a public venue, with I'm in and "2 of 4 going" shown as badges. A plan is **on** only once the minimum is reached; from an hour before, a **still in?** check — if it drops below the minimum, everyone is told before they leave. At the venue: **I'm here**, status chips ("on my way", "5 min late") instead of chat, and **Show your badge** to find the table.

COPY RULES

Nobody is ever told "X declined" or "X left". A passed proposal simply expires, a passed invitation reads "plan filled", a dropped seat changes "3 of 4 going" to "2 of 4 going" silently.

Motivation design: one mechanic per barrier

Inspired by behavioural activation — small, scheduled, concrete activities — as a *design principle*, not a clinical claim.

BARRIER	MECHANIC	STAGE
No energy to initiate	The app proposes, you only say yes. A proposal arrives with an activity you like, a time you're both usually around and a place halfway. One tap.	M0
"Will anyone come?"	Mutual by construction: a plan is confirmed only when both say yes; group plans show "2 of 4 going" and go ahead only at the minimum.	M0
Fear of being stood up	Reminder 2 h before; cancelling is one tap and reads "plan cancelled" — no reason, no blame; group plans add a still-in check.	M0 PUSH M1
Judgment	A vibe, never a face. You don't pick who takes your invitation, and nobody picks you by a photo.	M0
Talking is hard	Activity-first plans: board games, climbing, a walk, a quiz — something to do side by side.	M0
Day-of hesitation	The compass opens 15 minutes before — the last hundred metres become a game, not a search for a stranger.	M0
One meeting doesn't fix loneliness	Same again next week? after a plan; later circles — the same 3–4 people weekly for four weeks.	M0 CIRCLES V1
A small push · spam	Venue perk on check-in, paid by the venue; proposals expire on their own and declines are silent.	M1

Plan lifecycle

pass · expiry → it simply expires



pass → next person · they see "plan filled"

A plan stores who is meeting, never where they are. Group plans add "waiting for one more" before *confirmed*.

PLAN MATCHING

The Now loop's hard gates and score, plus an overlap in "when you're around" slots. The place fits the activity and sits roughly halfway ("7 min for you, 9 for them"); plans show the match %, so you know why it's worth your evening.

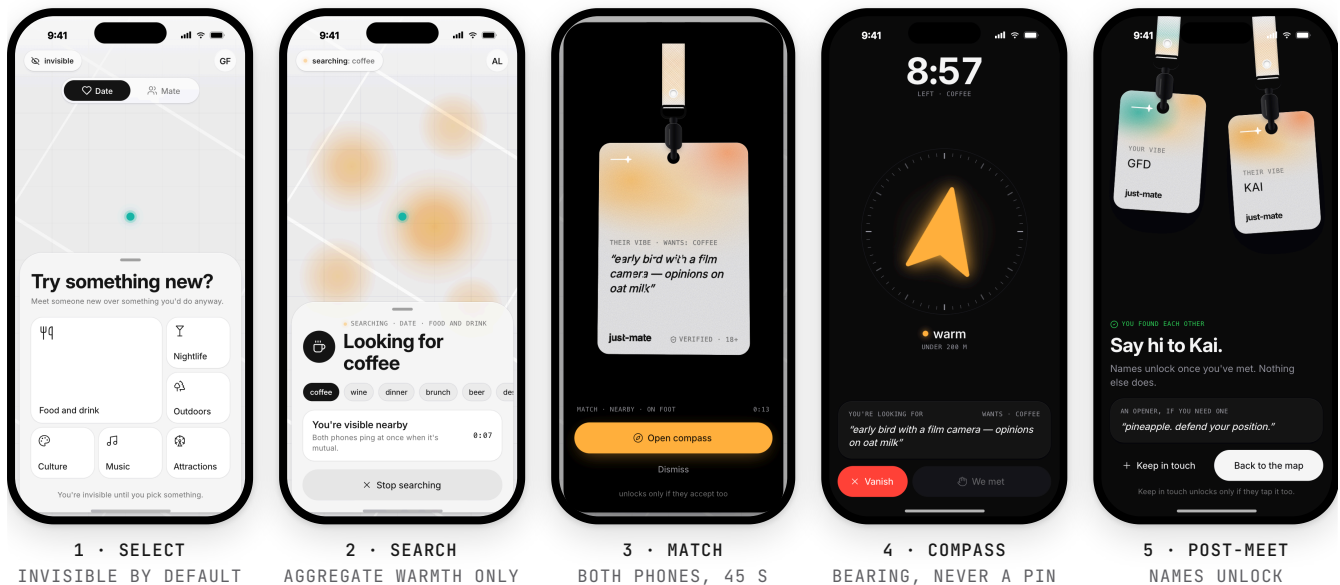
Availability. No calendars are read: JustMate knows when people are *usually around*, minus plans they already have. The tap confirms the rest — a proposal is on only when both accept; whoever takes an invitation picks the time.

PLACES

A curated list of public venues with kind, rating, opening hours and walk time; "places for you" are ranked by your interests ("you like board games"). For outdoor plans you can drop a spot on the map. **M1** Partner venues add slots and perks; dropped spots snap to public places.

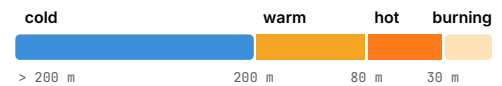
Consented serendipity

For when you're already out. happn proved proximity, Breeze proved skipping chat, live-map apps proved people love maps. Nobody assembled them: **real-time, mutual, faceless, on foot.**



- Pick what you're up for, deliberately.** Default is invisible. Choose Date or Mate, a category, one or more intents, then **Find people for {picks}**. Session-scoped, never ambient.
- Watch the map warm up.** A soft heat field shows where compatible people search for the same thing. No list, no browsing — scarcity of information is the feature.
- The ping.** Two compatible people, both searching, picks aligned, within walking range → both phones buzz at the same moment. Each sees only the other's badge: "their vibe · wants: beer" and their vibe line.
- Mutual accept.** The compass unlocks only if both tap **Open compass** within 45 s. A dismiss is never forwarded — the other side sees "offer expired", never "declined".
- The walk.** An arrow with hot/cold colour and escalating haptics; distance is a bucket. Ten-minute window.
- Meet. Talk.** At *burning*, **We met** unlocks first names, an opener if you need one, and a mutual **Keep in touch**. Then positions are discarded.
- Vanish** at any moment: one tap, no confirmation, the session is destroyed for both.

DISTANCE AS A GAME, DELIBERATELY IMPRECISE



The server sends **bearing + bucket**, never the partner's coordinates. **We met** unlocks only in **burning**.

PARAMETER	VALUE
Match radius (walk 5 / 10 / 15 min)	400 / 800 / 1200 m
Offer to accept	45 s
Compass session	10 min
Pair cooldown after any end	5 min
Search auto-stop	30 min
Position updates (search / session)	2 s / 1 s

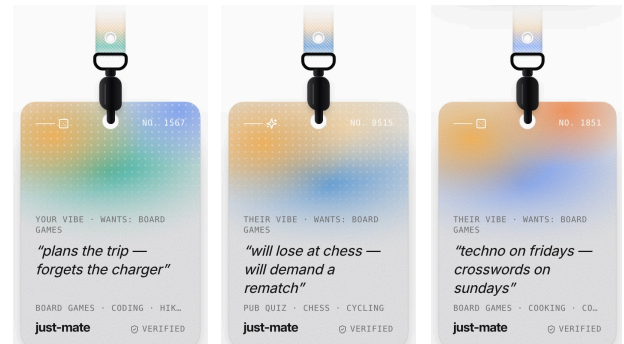
All thresholds are sent by the server in `config`, so tuning is a restart, not an app rebuild.

Not a profile-picture app

The whole funnel says it before you ever see the map. A two-minute onboarding turns your interests and a few answers into a **vibe on a lanyard badge** — the only thing anyone else ever sees.

Onboarding in about two minutes

- 1 Mode first.** Date ("Someone to fall for, a few streets away") or Mate ("People to grab a beer or a game with, right now"). Switchable on the map any time.
- 2 First name and interests.** The name stays hidden until you've met. At least three interests; each pick opens three related ones.
- 3 Four live questions.** An LLM writes each question from your previous answers, with four tap-able options — a conversation, not a form.
- 4 Your badge.** One wry line in two lowercase clauses — *"plans the trip — forgets the charger"*. Reroll until it fits, then **Keep this vibe**.
- 5 Who you're after.** Date: who, an age range, what you're looking for. Mate: anyone or same gender, group size, energy, and when you're usually around. "Used for matching only. Nobody sees your settings."
- 6 Taste (Date).** Swipe six *generated* sample faces — never real users. Only the traits they share are kept, as one line of words.
- 7 One selfie.** Described once (hair, face shape) for matching, then dropped — never stored, never shown. Date adds the blocking "I'm 18 or older" check.



Vibe badges. Colours, pattern, icon and serial number are designed deterministically from your interests and answers (badgeSeed) — no two alike. At a match you see their badge and the shared intent; nothing else.

WHAT A MATCH RECEIVES — THE WHOLE PAYLOAD

```
{ vibe: "techno on fridays — crosswords on sundays",
  interests: ["concerts", "pub quiz", "coding"],
  badgeSeed: 2915,
  tags: { verified: true, adult: true } }
```

No name, no answers, no score, no position.

Design principles

LAYOUT • BOLT

One surface over the map

The map is full-screen; a single surface morphs between shapes — auth, onboarding, select, search, plans, day-of, match, compass, post-meet, settings. One primary action at a time.

FEEL • APPLE FLUID INTERFACES

Instant, interruptible, tactile

Press feedback without latency, interruptible springs, translucent materials. Haptics are a language: [200,100,200] when a match lands, escalating pulses as the compass warms.

STONE • LIGHT, INK, NIGHT

Paper by day, night for the walk

A pale greyscale map with warm amber glow, an ink match card, a night compass. Red is reserved for Vanish. English, Polish and Slovak; reduce-motion, haptics and sound switches.

Safety is the architecture, not a feature

The jury's first question about any proximity app is "isn't this a stalker tool?" JustMate is built so the answer is **no, by construction** — and for women in particular, three layers sit on top: prove you're real, meet in public, remove someone from your world in one tap. **Every safety feature is free, forever.**



Mutual by construction

Both are notified, both must accept before anything unlocks. A dismiss is never forwarded — the other side only sees "offer expired".



A bearing, never a position

The server sends a direction and a distance bucket — never the partner's coordinates. "No pins" is enforced in the protocol, not in the UI.



Zones, not pins

The map shows aggregate warmth per zone; zones with fewer than 3 searchers stay dark (k-anonymity), so a glow never points at one person.



Invisible by default

Nothing is discoverable until you start a search or join a plan. Search auto-stops after 30 minutes (on by default). Nobody can be searched or browsed — ever.



Ephemeral and time-boxed

45-second offers, 10-minute compass sessions, a 5-minute pair cooldown. Positions live in memory per socket and are dropped when a session ends.



Vanish

One tap, no confirmation: the session is destroyed for both, instantly. It looks identical whichever side pressed it.

What we keep, what we hold briefly, what we never touch

STORED • POSTGRESQL

- Account: email and a password hash (Better Auth), sign-in sessions
- Profile: mode, first name, age, gender, interests, answers, vibe line, preferences, settings
- Text-only matching lines: appearance traits, taste traits, character — never shown to anyone

MEMORY ONLY • DISCARDED

- The current search: mode, category, intents
- Your last position — only while searching or in a session
- Offers, compass sessions, cooldowns
- Plans: who is going, never where they are

NEVER

- Any photo of you (the selfie is described once, then dropped)
- Location history, or logs of positions
- Messages — there is no chat
- Mood or health data
- Anyone's coordinates on your phone

Your data rights are endpoints, not a support ticket: `GET /api/account/export` returns everything stored about you; `DELETE /api/account` removes the user, profile, sessions and sign-in methods and closes the live socket.

Threat model

THREAT	COUNTERMEASURE
Stalker follows a specific person	No identity, no search, no browsing; zones not pins; aggregate glow with k-anonymity; nothing discoverable when search is off; plans are proposed by the app, never browsable by person.
Compass abused to locate someone	Unlocks only after mutual accept; reveals a bearing, never a map position; 10-minute TTL; symmetric — the partner knows you're walking too. M1 meeting point first.
Plan used to reach a specific person	Nobody picks a person: proposals come from the model, invitations are offered by the app ("you don't pick who"); the other side sees a vibe, a place and a time; a pass is never shown. M1 people who met and didn't both keep in touch aren't grouped again for 30 days.
Harassment · hostile profiles	No chat, so no DM channel. Every saved profile passes OpenAI moderation; a flagged person silently drops out of matches, zones and plans. M1 block and report remove someone for good.
Serial abuser cycling matches	M1 At most 5 compass sessions a day and 3 open plans; repeated vanishes or reports from <i>different</i> people flag the account.
Fake or bot profiles · ban evasion	Selfie liveness check (simulated in the build); M1 one account per phone number, optional ID check, bans bound to phone and device.
Weaponised false reports	M1 Auto-pause needs two independent reporters; verified reporters weigh more; a human reviews before any ban; appeal path.
Data breach · subpoena	Positions are never persisted; plans hold who, never where. There is nothing to leak.
Minors	Every profile states an age (16+). Date requires an adult, checked server-side. In Mate, a non-adult is only ever matched or grouped with another non-adult . M1 age assurance.

Verification ladder

LEVEL	CHECK	STAGE
1 · Account	Email + password or magic link	M0
2 · Phone	SMS code, one account per number (~€0.05–0.10)	M1
3 · Liveness	One selfie, deleted after the check → "verified"	SIM. M1
4 · ID	Document + selfie via an ID provider (~€0.70) → "ID verified"; required to host plans	M1

The user never pays for a level. "Only match me with verified people" is a free filter.

Block, report and meeting in public

- **Block** from post-meet, a plan, or **Recent** (last 7 days, badges only): instant, permanent, two-way.
 - **Report = block + flag**, with an optional note — the only free text in the app, read only by reviewers.
 - **Auto-pause** after two independent reports in 30 days, until a human reviews.
 - **Meeting point first** (default in Date): the compass leads both to a public spot, not to each other. **Women-only plans** and a **trusted contact** share.
- M1** All of the above are production path; in the build, Vanish is live.

FOR WOMEN SPECIFICALLY — TODAY, IN THE BUILD

She is invisible unless she starts a search or joins a plan. She can dismiss any match invisibly — on the other side it simply "expires". Vanish is one tap and instant. Her first name appears only after she has met someone in person.

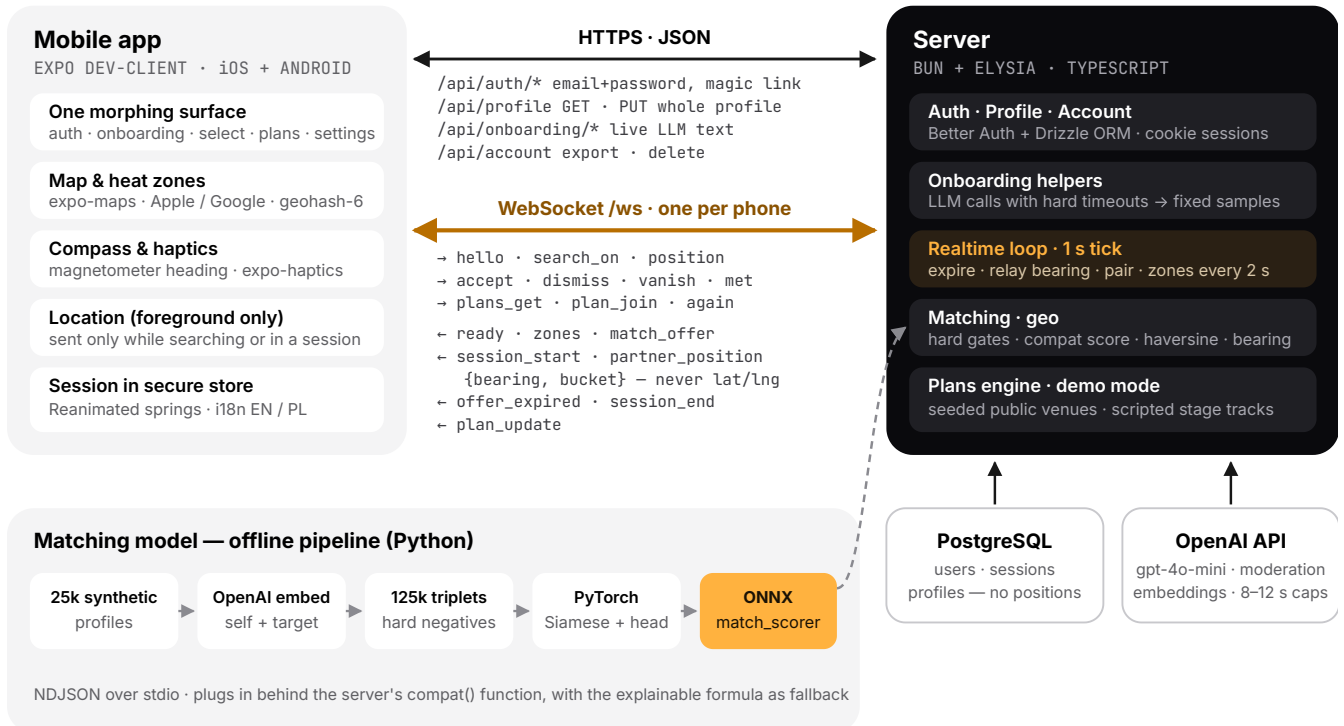
GDPR / RODO POSTURE

Location is personal data, so it is processed only inside an explicit, session-scoped consent and never stored. No special-category data. The photo goes to the LLM once and is not stored or logged. Production path: a DPIA before launch; the demo runs on test data only.

Two processes, one contract

A native app and a Bun server speak one typed protocol: HTTP for accounts, the profile and onboarding; **one WebSocket per phone** for everything live. The contract was written first — `PROTOCOL.md` → `@justmate/protocol` → code — so mobile and backend were built in parallel against it.

`@justmate/protocol` — one TypeScript package: wire types, validators (`parseProfile`), categories, `badgeSeed` · imported by app and server



System overview. The only coordinates a phone ever receives are venues. Position data flows phone → server memory → (as a bearing and a bucket) → the one matched partner, and is dropped when the session ends.

STACK · MOBILE

Expo SDK 57, React Native 0.86, Expo Router, Reanimated 4, Gesture Handler, Bottom Sheet, **expo-maps** (native, hence a dev-client build — not Expo Go), expo-location, expo-sensors, expo-haptics, expo-secure-store, Better Auth Expo client, i18next.

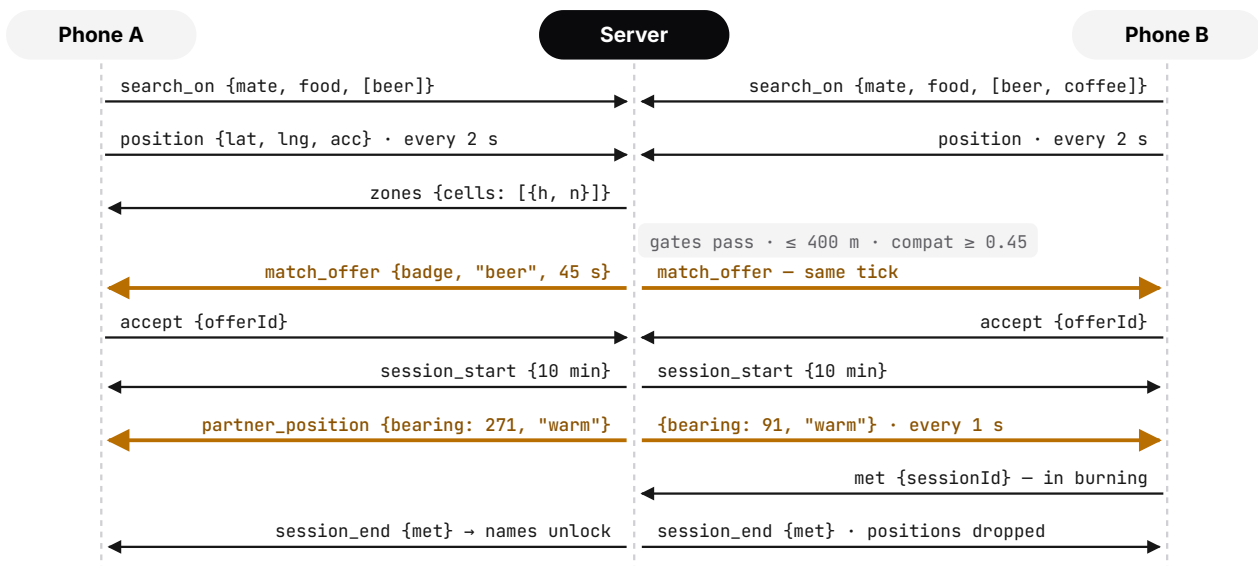
STACK · SERVER

Bun, Elysia (HTTP + ws), TypeBox validation, **Better Auth** (email + password, magic link, password reset over SMTP), **Drizzle ORM** with versioned migrations on PostgreSQL, OpenAI API over plain fetch (gpt-4o-mini, moderation, embeddings), i18next for server-side strings.

STACK · ML

Python with uv, PyTorch, scikit-learn, ONNX Runtime, OpenAI text-embedding-3-small. A Jupyter evaluation notebook with ROC/PR curves, calibration, top-K retrieval and t-SNE of the learned space.

A match, end to end



Protocol rules that carry the product

- **The server is authoritative.** Clients only *request* (accept, vanish) and *report* (position).
- **A dismiss is never forwarded.** Expiry, dismiss, disconnect and a replaced search all look the same to the other side: `offer_expired`.
- **Bearing and bucket only**, computed server-side; exact metres go only to demo sockets for tuning and are never rendered.
- **One offer or one session per user.** Greedy pairing by score each tick; ties go to the shorter distance.
- **Socket close = search off.** Nothing live survives a reconnect — there is nothing to reconnect to.
- **Relative times** (`expiresInMs`), so phone clock skew never matters; unknown events are ignored, never fatal.

Built for a live stage

GPS inside an arena is unreliable, and nobody waits until Thursday on stage. A **demo mode** (`?demo=a|b`, dev builds only) feeds scripted positions through the *identical* pipeline: phone B converges on A over 40 s (cold → warm → hot → burning). The track is defined relative to the stage, because the compass arrow uses the phone's real magnetometer — a mismatched script would point into the audience.

Demo sockets pair only with each other; nine server-side **ghosts** add map density and **never match**. A dev time-skip makes a plan start two minutes after it is proposed. All of it is labelled in the README's real-vs-canned table.

EVERY 1-SECOND TICK, THE SERVER

1. expires offers (45 s) and sessions (10 min), and puts the pair in cooldown;
2. relays `partner_position` as bearing + bucket to both members of each session;
3. pairs searchers: all pairs passing the gates, highest score first, at most one offer per user;
4. every 2 s, sends each searcher their `zones`, omitting cells under $k = 3$.

CONFIG · SENT ONCE IN READY

```

offerTtlMs 45000      sessionTtlMs 600000
pairCooldownMs 300000 autoStopMs 1800000
walkRadiusM {5:400, 10:800, 15:1200}
buckets {warm:200, hot:80, burning:30}
zonePrecision 6      kAnonymity 3
  
```


Explainable first, learned second

The demo runs on a transparent score we can put on a slide and defend in Q&A. AI does two jobs: an LLM interviews you and writes your vibe, and a learned model is trained to replace the formula once real meetings exist to learn from.

Hard gates — never learned, never scored

Whatever scores a pair, it only sees pairs that already passed every rule below:

- Both searching, same mode and category, **at least one shared intent** (the context of the match: "this is a beer match")
- Within **walking distance**: haversine $\leq$ the shorter of both walk settings (400 / 800 / 1200 m)
- Age and safety**: adult flag equal on both sides; Date requires adults; each side's gender preference and age range accept the other — checked both ways
- Pair not in cooldown; neither has an open offer or session; not self; not a ghost; demo sockets only with demo sockets

The explainable score

```
compat = 0.7 × Jaccard(interests) + 0.3 × min(1,
|shared intents|)
match ⇔ gates pass ∧ compat ≥ 0.45
```

Mode and intents gate, interests score. In Now the score is never shown — the badge and the shared intent carry the moment; a plan proposal shows it as a match percentage.

WHY DISTANCE, NOT ZONE

Zones are a *display* unit (geohash-6, $\sim 1.2 \times 0.6$ km in Kraków). Matching on "same cell" would pair people up to **1.35 km** apart — more than the **~830 m** a person walks in the 10-minute window — and would never pair two people 50 m apart on either side of a cell boundary.

400 m $\approx$ 5 minutes on foot, leaving half the window to find each other.

ZONES ON THE MAP

Each searcher gets, every 2 s, the count of compatible searchers per geohash-6 cell within 2 km. Cells with fewer than 3 are omitted, so a glow can never be followed to *the only* person in it.

WORKED EXAMPLE · MATE · FOOD AND DRINK

A picks beer · coffee

B picks beer

A interests board games · climbing · chess · coding · pub quiz

B interests board games · chess · coding · concerts · cycling

Jaccard 3 shared / 7 distinct = 0.43

compat $0.7 \times 0.43 + 0.3 \times 1 = 0.60 \geq 0.45 \rightarrow$ a beer match, if they are within 400 m

The LLM in onboarding

ENDPOINT	WHAT IT WRITES	TIMEOUT
question	The next onboarding question, written from your previous answers, with exactly four short options	12 s
vibe	Your badge line: two lowercase clauses, ≤ 60 characters, no names or emoji, never one you rejected	10 s
related	Three interests close to the one you just picked	8 s
character	Five "trait — concrete detail" lines for matching; no looks, age, names or places; never sent to a match	10 s
taste · appearance	Traits shared by the sample photos you liked; visible hair and face features from your selfie — never age, ethnicity, emotion or identity. The photo is not stored or logged	8 s · 12 s

Model: OpenAI `gpt-4o-mini` with strict JSON-schema output (overridable via `OPENAI_MODEL`). On a missing key, a timeout, an API error or output that breaks the rules, the server returns a fixed sample (`source: "sample"`), so the app never waits on an error path.

The learned compatibility model

An **asymmetric Siamese network**. Each profile is embedded twice with OpenAI `text-embedding-3-small` (1536-d): once as *who I am* (self) and once as *who I'm looking for* (target). A shared encoder projects both into a 128-d compatibility space; a match head scores the pair.

Shared encoder	1536 $\rightarrow$ 256 $\rightarrow$ 128 · LayerNorm + ReLU · L2-normalised
Match head	[$ z_a - z_b $, $z_a \odot z_b$, $\cos$] · 257 $\rightarrow$ 128 $\rightarrow$ 32 $\rightarrow$ 1
Direction	$\text{head}(\text{target}_A, \text{self}_B) \rightarrow \text{sigmoid}$
Pair score	$\text{dir}(A \rightarrow B) + \text{dir}(B \rightarrow A) - \text{symmetric}$, range 0-2
Loss	triplet (margin 1.0) + 0.5 · BCE
Training	cosine LR · input noise σ 0.01 · hard negatives re-mined every 2 epochs · 80/20 split by anchor

Why two heads: raw embeddings encode general similarity, not compatibility. The triplet loss reshapes the space; the head learns the decision boundary and handles calibration.

PRODUCTION PATH Retrain on real outcomes — mutual accept + met $\rightarrow$ 1, dismissed or vanished $\rightarrow$ 0, **same again** as the strongest positive; per-user vectors cached in PostgreSQL with pgvector; threshold calibrated on held-out real interactions (target FPR $\leq 5\%$, TPR $\geq 80\%$).

0.765

validation ROC-AUC

0.753

F1 at the best symmetric threshold (0.85)

25k

synthetic profiles with descriptions

125k

training triplets (5 per anchor)

Honest limit. Before launch there are no real meetings to learn from, so the labels are synthetic and rule-based. The model is exported to ONNX and served by a standalone scorer (NDJSON over stdio) that plugs in behind the same `compat()` function, with the formula as fallback — **the demo runs on the explainable score**. We never call it "AI matching" on stage.

Evaluated in a regenerable notebook: ROC and PR curves, confusion matrix, accuracy per hard-negative bucket, a reliability (calibration) curve, top-K retrieval over all 25k profiles, and a t-SNE of the learned space.

Protocol first, tests that read like the spec

Built in 24 hours by a team working in parallel. What made that possible: the contract was written before the code, a mock server replayed it so the app never waited on the backend, and the safety rules live in tests, not just in documents.

The repository

PATH	WHAT LIVES THERE
mobile/	Expo dev-client app: auth, onboarding (Date / Mate), map with categories and heat, Plans, match card, compass, post-meet, settings — one morphing surface
server/	Bun + Elysia + Drizzle: auth, profile, live onboarding text, zones, matching loop, compass relay, Plans; a mock that replays the protocol's happy path
packages/protocol	The wire types and validators shared by app and server
ml/	Synthetic data → embeddings → triplets → training → ONNX → evaluation notebook → scorer
docs/	Product definition, screens and copy, design system, protocol, ML spec, demo script, build plan

Quality gates

- 118 automated server tests (Bun test) across pairing, offers, sessions, zones, matching rules, Plans, profile validation, moderation, embeddings, onboarding fallbacks, the WebSocket handshake and localization.
- Biome lint and TypeScript typecheck on every package; `bun run lint && bun run typecheck && bun run test` before every push.
- Drizzle owns the schema with versioned SQL migrations; Better Auth owns its tables.
- GitHub Actions: conventional-commit version bumps and release-on-tag; EAS builds of the dev client for both demo phones.

TEST NAMES, VERBATIM

- "partner_position carries bearing and bucket, never coordinates"
- "an adult is never offered a non-adult"
- "a dismiss reaches both sides as a plain expiry and both keep searching"
- "cells with fewer than k compatible searchers stay hidden; the recipient never counts"
- "the partner card carries the badge and nothing else"
- "the match radius is the shorter of both walks"
- "a dismissed pair is not offered again until the cooldown ends"
- "the demo pair runs the whole happy path on the scripted track"
- "the selfie is not described without a model"

RUN IT

```
bun install
cp server/.env.example server/.env
bun --cwd server db:migrate
bun run dev:server
bun run dev:mock
bun --cwd mobile ios
```

Server on :3000 (WebSocket at /ws), protocol mock on :3001. Needs Bun ≥ 1.3, PostgreSQL and Xcode or Android Studio. Without an LLM key, onboarding uses fixed samples.

Venues pay. People meet for free.

Four principles: meeting is never paywalled; safety is never paywalled; those who earn from people going out — venues and events — pay most; premium sells convenience and reach, not access.

Revenue streams

STREAM	HOW IT WORKS
Partner venues PRIMARY	Cafés, board-game bars, climbing gyms, cinemas list plan slots for ~€39 a month and fund perks on check-in. They fill off-peak tables; their slots seed density. Partners see counts and check-ins, never identities.
Prepaid first round	An optional €5 voucher at partner venues, sold as a €20 wallet top-up; JustMate keeps 15%. Doubles as a commitment device — prepaid people show up.
JustMate+	€4.99 a month: host your own group plans (ID-verified hosts, public venues), travel mode, private plans for existing friends. Proposals and 1:1 invitations stay free.
Events & festivals	Licensed official zones and plans for an event, €500–2,000 each.
Later: B2B2C	Employers onboarding relocated staff, universities onboarding international students, city wellbeing programmes.

REJECTED ON PURPOSE

"Five free connections, then pay"

It paywalls the north-star event and charges lonely people for the one thing they came for. It rewards rationing meetings — the swipe-app model we position against. Limits exist in JustMate, but for safety, not revenue.

NEVER

Ads in the meeting flow. Selling attention. Selling data. A paid safety feature. JustMate's success metric *decreases* phone usage, so attention is not our product.

Back of the envelope: one city, 5,000 monthly active users

MONTHLY COST	ASSUMPTION	€	REVENUE	ASSUMPTION	€
Hosting	API, PostgreSQL, backups	100	Venues	25 × €39	975
LLM	~1,000 new × ~€0.01, openers	20	Prepaid round	900 × (€0.75 – card fee)	380
Phone checks	1,000 × ~€0.07	70	JustMate+	150 × €4.99, net of VAT, fees	520
ID checks	40% of new users × ~\$0.80	300	Events	1 festival a quarter	330
Moderation	20 h × €15	300	Total		~2,200
Stores, email	Apple fee, email plan	30			
Total		~820			

Readout: ~21 partner venues × €39 cover a city's running costs on their own (820 ÷ 39); everything else is margin toward a team. All figures are labelled estimates to be replaced with real quotes.

Plans lower the density bar

The 2012 proximity apps died of empty rooms. JustMate's answer to cold start is structural: **a new city opens with plans; the Now loop lights up as density grows.**

NOW • NEEDS PEOPLE IN THE SAME MINUTE

~9

people searching at any evening minute in a district of 500 active users who each search 30 minutes a week — split across two modes and a dozen intents. A Now match is rare.

PLAN • NEEDS PEOPLE OVER TWO DAYS

~16

people per intent over the next 48 hours if 40% of the same 500 are open to plans. A 1:1 plan needs one of them; a group of four fills easily.

Go-to-market

- **Single-density launches:** one campus, one festival, one district — anywhere the glow is populated from day one.
- **Event mode:** festivals and conferences pre-seed official zones and plans for a known, opt-in crowd.
- **Venue partners:** board-game cafés, breweries and climbing gyms host plan slots — density and revenue at once.
- **Newcomer channels:** campus societies, expat communities, HR onboarding, international-student offices.

REFERRAL CHANNEL • PSYCHOLOGISTS AND COUNSELLORS M1

Referral, not treatment

Psychologists, counsellors, student counselling centres and GPs see lonely clients every week, and often suggest exactly what Plans make easy: small, concrete, scheduled activities. JustMate is a next step to recommend between sessions — the way they'd recommend a climbing club. Referring people to community activity is established practice: in England, GPs referred 1.3 million people to social prescribing in 2023 (UCL, 2025).

Without health data. A referral kit (leaflet + a QR code per practice); the client signs up like anyone else — no "patient" flag, nothing reported back. Installs are counted per practice and never stored on a profile, because "referred by a therapist" is health data (GDPR Art. 9). No health claims and no clinical integration keep JustMate outside the EU MDR. Pilot: 3–5 practices and a university counselling centre in Kraków.

Competitive positioning

	JUSTMATE	BLINDMATE	TIMELEFT	HAPPN	TINDER	MEETUP
A concrete plan that gets you out	✓ app-made, nearby	✗	✓ weekly slots	✗	✗	needs organiser
Real-time proximity	✓ live zones	✗ region	✗	retrospective	✗	✗
Faceless	✓ always	until you chat	little info	✗	✗	n/a
Ends in person, not in chat	✓	✗ chat first	✓	✗	✗	partly
Walkable (5–15 min)	✓	✗ 20–50 km	✗ city-wide	partly	✗	✗
Works without a friend circle	✓	✗ friends swipe	✓	✓	✓	✓
Friendship, not only dating	✓ Mate + Date	✗ dating	✓	✗	✗	✓
Consent architecture	✓ mutual + zones	friends decide	partly	partly	✗	n/a
Core free to the user	✓	freemium	✗ per seat	freemium	freemium	✓ to attend

Closest in spirit: BlindMate

BlindMate (Germany, since 2021; 1M+ downloads, 4.7★ on the App Store, mostly German-speaking countries) also refuses the photo audition. Your **friends** write your profile and swipe for you; when both sides' friends agree, a "blind chat" opens and the photos unlock as you write. Same instinct as ours — a different answer at every step after it.

	BLINDMATE	JUSTMATE
Who finds the match	Your friends, swiping for you	A compatibility model. No circle needed — for a newcomer like Tomek, having no one to ask <i>is</i> the problem
Photos	Hidden at first, unlocked during the chat — the audition is postponed	Never shown. The vibe badge is all anyone sees, before and after you meet — the audition is removed
After a match	A chat; meeting is up to you	No chat: a plan with a time and a public place halfway, or a ping and a compass within a 5–15-minute walk
Distance	Region, in 20 / 50 km steps	Walking distance (400–1,200 m), live or as a plan halfway
Purpose	Dating	Friendship and activity partners first — including sport (Mate) — dating second (Date)
Health angle	None claimed	Built around loneliness: meet, move, look up; a referral channel through psychologists and GPs (ch. 01, 12)
Who pays	Free with in-app purchases	Free to the user; venues pay (ch. 11)

Where BlindMate is ahead — and what we take from it. Real traction, and the warmth of being described by people who know you: a friend's word is social proof. Our answer to trust today is the vibe badge and the verification ladder (ch. 07). A friend's vouch would fit all three rules — no faces, no chat, no pins — and is a candidate for later, not a promise.

Wedge: consented serendipity, plus plans you only have to say yes to — faceless for good (not just until the chat), no chat at all, walkable, open to people with no circle yet — with Timeleft's proof that strangers show up when the plan is concrete.

Measured on the street, not the screen

NORTH STAR

Completed real-world meetings per active user per week

Plan and Now combined.

MISSION METRIC • LONELINESS

Repeat meetings

Share of active users who met the same person or group at least twice in 30 days. Repetition is how friendship starts.

ANTI-METRIC, WATCHED ON PURPOSE

Time in app should not grow

Engagement is measured on the street. The phone's job ends where the conversation starts.

Activation	Share of new users who attend a first plan or meeting within 7 days of signing up
Plan health	Proposal acceptance (proposed → both in); invitation take rate (sent → someone's in → confirmed); group fill rate; show-up rate ≥ 80% of confirmed people; stood-up rate (checked in, nobody else came) under 2%
Safety guardrails	Vanish rate under 10%; reports per 1,000 meetings; median time to review a report under 24 h; notification opt-out rate
Movement	Share of meetings that are sport or outdoors; minutes walked to meet — aggregate only, never a track
Outcome (research only)	Opt-in, anonymous 3-item UCLA Loneliness Scale and WHO-5 Well-Being Index at sign-up and after 8 weeks, with a research partner — aggregated, never stored on a profile

Roadmap

STAGE	SCOPE
M0 HackYeah	Native app (Date / Mate, categories, vibe badge, compass, haptics, 18+ gate, post-meet), WebSocket server with distance-gated matching and TTLs, explainable scoring, demo mode; Plans (Date and Mate: model proposals and 1:1 invitations with several times, mutual confirm, reminder, plan-time compass; group plans for 3–4 in Mate with I'm in, go-ahead minimum, still-in, status chips, show your badge; same again); help row with support lines; ML pipeline trained on synthetic data
M1 Production MVP	Remote push (background search, proposals and invitations while the app is closed); referral kit for psychologists and counsellors with a Kraków pilot; block, report, auto-pause and Recent; phone and ID verification; verified-only and women-only filters; meeting point first; trusted contact; rate limits; partner venues and perks; prepaid first round; verified-only Date plans; age assurance; DPIA; end-to-end encrypted positions between paired sessions; model v1 on real outcomes
V1	Circles (same 3–4 people weekly for four weeks), user-hosted group plans, opt-in calendar free/busy (busy blocks only, never titles or details) so plans skip times you're booked, JustMate+ (travel mode, private plans), a private "brave log" of badges you've met — it only grows, nothing to lose
V2	Venue and event platform (official zones, analytics), city heat events, social-prescribing partnerships (GP link workers, university counselling, employee assistance programmes), employer and university programmes, partner channels for older adults

Deliberately excluded, forever: chat, photo profiles, browsing people, followers and likes, feeds, ads in the meeting flow, paywalled connections, streaks that reset, "you haven't gone out" nudges, health claims.

What's real, what's canned

Rule for every sentence in this document, the deck and the description: it describes what the demo does, or it is labelled production path.

REAL IN THE BUILD

- Accounts, profiles, onboarding with live LLM text (fixed samples without a key)
- Zone glow from live positions, with k-anonymity
- Mutual match delivered live to both phones over WebSocket, with haptics
- Explainable compatibility scoring and every hard gate, including age segregation
- Compass from the real magnetometer; bearing + bucket relay; Vanish; We met; post-meet
- Plans: proposals from real profiles, accepted on both phones; invitations with several times; the confirmed plan with its compass; group plans with I'm in and the go-ahead minimum
- Account export and delete

CANNED, AND LABELLED AS SUCH

- Demo-mode positions (indoor GPS) on a scripted track relative to the stage
- Nine ghost wanderers adding map density — they never match or join plans
- A demo time-skip that starts a plan two minutes after it is proposed
- The seeded list of public venues; canned attendee badges, if shown
- The selfie liveness check (simulated; production path)
- The learned model: trained on synthetic labels, not in the live match loop

AI-use disclosure

AI tools were used for ideation support and boilerplate; the core idea, mechanic and solution are the team's work. The product itself uses AI: an LLM (OpenAI `gpt-4o-mini`) interviews the user and writes their vibe line and matching traits; OpenAI moderation screens every saved profile; OpenAI embeddings encode profiles for the learned model. Matching and plan proposals use an explainable compatibility score. A compatibility model was trained during the event on synthetic profiles (OpenAI `text-embedding-3-small` embeddings, PyTorch).

Credits & licences

Base maps by Apple Maps (iOS) and Google Maps (Android), attribution shown by the native map. Icons: Lucide. Type: Inter. Taste-swipe sample faces are generated, never real users. The demo runs on test data only.

Sources

1. WHO Commission on Social Connection, *From loneliness to social connection*, and WHO news release, 30 June 2025 — prevalence, ~871,000 deaths a year, 2× depression, stroke, heart disease, diabetes, cognitive decline; social connection as a public-health priority.
2. Holt-Lunstad et al., *PLoS Medicine*, 2010 — 148 studies: 50% higher odds of survival.
3. WHO, 26 June 2024 — 31% of adults insufficiently active.
4. Hall, J. *Soc. Pers. Relat.*, 2018 — ~50 h to a casual friend.
5. UCL, September 2025 — 1.3 million GP social-prescribing referrals in England in 2023.
6. BlindMate: blindmate.de, App Store listing (Oct 2026), netzwelt.de (2024).
7. EU MDR 2017/745; GDPR 2016/679, Art. 9. Pricing assumptions are the team's estimates.

TEAM

Arthur Kozubov · Jozef Zvalo · Andrej Zak ·
Nikita Orlov · Serhii Vielkin

FIND EVERYTHING

Code and docs: github.com/uteg-labs/just-mate
Start with `README.md`, then `docs/PRODUCT.md` and
`docs/PROTOCOL.md`.

Not therapy. A reason to go out — and someone waiting when you get there. **Meet for real.**